# 劉晉安 Chin-An, Liu

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## *SUMMARY*

Master of Computer Science graduate, specialising in Data Science and AI. Focused on **MLOps and full-stack AI platform deployment**. Experienced in optimizing model inference (75% latency reduction) and managing end-to-end model lifecycles. Dedicated to enhance **manufacturing productivity and system scalability** through robust AI solutions and collaboration.

## *EDUCATION*

**University of Sydney** — Sydney, Australia

Master of Computer Science — Jul. 2024 ~ Nov. 2025

**WAM:** 76, **GPA:** 3.4

**National Pingtung University** — Pingtung, Taiwan

Bachelor of Computer Science and Artificial Intelligence — Sep. 2020 ~ Jun. 2024

**WAM:** 87.1, **GPA:** 3.67

## *SKILLS*

**ML & DL:** CNN, YOLO, Transformer, PyTorch, TensorFlow, Scikit-learn, Computer Vision (OpenCV), Prompt Engineering

**MLOps & Full stack Development:** Docker, CI/CD Pipelines, Git, React, Node.js, SQL

**Programming Languages:** Python, C/C++, C#, Java, R

**Languages:** English (IELTS Academic 6.5), Mandarin (Native)

## *WORK EXPERIENCE*

**Multimedia Research Assistant** — Pingtung, Taiwan

National Pingtung University — Oct. 2023 ~ Jul. 2024

- Optimised Reversible Data Hiding (RDH) algorithms to enhance data embedding capacity (EC) by 13%, ensuring image data integrity.
- Synthesized State-of-the-Art (SOTA) research from multiple journals to identify and evaluate optimal methodologies.
- Established a standardized literature review process that improved the precision of experiment reproduction and validation.

## *AWARDS & CERTIFICATES*

Google AI Professional Certificate, App Building, Data Analysis — 2026

書卷獎 — 2022, 2023

## *PROJECTS*

**Master's Capstone Project** – Automated Detection of Pollinator–Flower Interactions from Citizen Science Images — 2025

- Led a 7-member team by defining project direction, proposal and system development. Tracked progress and optimized team collaboration processes in weekly meetings, ensuring smooth workflows, and alignment with client requirements.
- Developed and productionized an Automated Floral Feature Recognition System and fine-tuned YOLO and a multi-head Vision Transformer for ROI detection and feature extraction. Achieved over 90% accuracy, providing real-time automated analysis.
- Optimized AI inference pipelines within a Docker-based Full-stack environment, successfully reducing end-to-end latency from 3s to 750ms (75% reduction) to support high frequency real-time analysis and improve system throughput and scalability.

**Bachelor's Capstone Project** – 結合深度學習與影像加密技術在資訊通訊上安全機制之設計與實作 — 2023

- Led 3-member team, organising code reviews and progress meetings to resolve efficient problem-solving.
- Developed a Biometric Authentication System integrating Face Recognition, Visual Speech Recognition and RDH algorithms. Improved verification accuracy from 79% to 92% to meet real-time financial verification requirements.
- Established modularized development workflows and documentation standards, ensuring encryption algorithms complied with communication security protocols, providing security and scalability guarantees for large-scale system deployment.